



EXPLOITS UNIVERSITY

BSc INFORMATION COMMUNICATION AND TECHNOLOGY

BICT 3202 · OBJECT-ORIENTED ANALYSIS AND DESIGN

WEEK 5

Assignment: Object-Oriented Analysis — Object Modelling I

PROGRAMME

BSc Information Communication and Technology

COURSE CODE / YEAR

BICT 3202 · Year 3

LECTURER

Francis Fweta — ICT Lecturer

DELIVERY

2h Lecture · 1h Tutorial · 1h Practical · 10h Independent Learning

Prepared by Francis Fweta · Exploits University · Week 5 of 16



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ASSIGNMENT INSTRUCTIONS

- Answer ALL questions in Sections A, B and C.
- Read the Week 5 module and your lecture notes before attempting each question.
- Draw diagrams neatly (by hand or using a UML/CASE tool) with correct UML notation.
- For every association, show a meaningful name and multiplicities where required.
- Cite OMG UML 2.5.1 where relevant.
- Submit by the date given by your lecturer. Total: 50 marks.

Section A — Short-Answer Questions (20 marks)

Question A1

Define Object-Oriented Analysis. [2 marks]

Question A2

Differentiate between a class and an object, giving one example of each. [3 marks]

Question A3

What is an attribute? Give two attributes of the class Course. [2 marks]

Question A4

What is an operation? Give two operations of the class Student. [2 marks]

Question A5

Differentiate between a link and an association. [2 marks]

Question A6

What is multiplicity? Explain the meaning of 0..1, 1, 0..* and 1..*. [3 marks]

Question A7

Define generalization and specialization. [2 marks]

Question A8

Explain the 'is-a' test and use it to decide whether Student $\square$ Course is a generalization. [2 marks]

Question A9

What is a CASE tool, and why can it not replace an analyst? [2 marks]

Section B — Application Questions (20 marks)

Question B1 (20 marks)

'A university has students and lecturers. Each student has a student ID, name, programme and year of study. Each lecturer has a staff ID, name and department. Students register for courses. Each course has a course code, title and credit hours. Lecturers teach courses. A student may register for many courses, while each course may have many students.'

- (a) Identify the candidate classes. [3 marks]
- (b) Identify at least two attributes for each class. [4 marks]
- (c) Identify one or two operations that could reasonably belong to each class. [3 marks]
- (d) Draw a UML class diagram showing the associations, with meaningful names. [5 marks]
- (e) Add appropriate multiplicities to the associations. [3 marks]
- (f) State whether generalization is appropriate, and justify your answer. [2 marks]

Section C — Case Study (10 marks)

'A university wants an Accommodation Management System to manage students, rooms, hostels, accommodation applications, payments and wardens. Students apply for rooms. A hostel contains several rooms. A student can occupy one room at a time. A warden manages a hostel. Students make accommodation payments.'

As an object-oriented analyst:

- (a) Identify the candidate classes (at least six). [3 marks]
- (b) Provide at least two attributes for each major class. [3 marks]
- (c) Draw the associations between the classes and add multiplicities. [4 marks]

Marking Scheme

Section	Content	Marks
A	Short-answer questions (9 questions)	20
B	Application — class diagram (parts a-f)	20
C	Case study — accommodation system	10
	TOTAL	50

Submission & Academic Integrity

- Submit a typed document (or neatly handwritten, as directed by your lecturer).
- This assignment is individual work. Plagiarism or copying will attract a penalty under university regulations.



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- Diagrams must use correct UML notation — marks are awarded for notation, association names and multiplicities.
 - Cite any sources you consult using the format used in the Week 5 module references.